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Zuoyan Zhang

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Education

Hunan University Sep. 2024 – Present
Ph.D. in Computer Science and Technology, College of Computer Science and Electronic Engineering Changsha, China

- Advisor: Jie Zhao
- Research Interests: AI compilers; systems for large-model training and inference

University of Lisbon Aug. 2026 – Present
Visiting Ph.D. Student in Computer Science and Technology, Instituto Superior Técnico Lisbon, Portugal

- Advisor: Nuno Lopes
- Research Interests: LLVM compiler optimizations; large-model inference acceleration

Information Engineering University Sep. 2021 – Jun. 2024
M.Eng. in Computer Technology, School of Cybersecurity Zhengzhou, China

- Advisors: Jinchen Xu and Jie Zhao
- Laboratory: State Key Laboratory of Mathematical Engineering and Advanced Computing
- Research Interests: floating-point error analysis; polyhedral compilation

Henan University of Technology Sep. 2017 – Jun. 2021
B.Eng. in Computer Science and Technology, College of Information Science and Engineering Zhengzhou, China

Experience

Huawei Technologies Co., Ltd. Jul. 2025 – Mar. 2026
Compiler Development Intern

- Implemented automated parallel-configuration search in MindSpeed, reducing search overhead by 5.6× while achieving 1.04× the performance of expert-tuned configurations.
- Scheduled communication operators at the computation-graph level to overlap communication with computation, achieving 1.14× the performance of SimpleFSDP and 1.10× that of DeepCompile.
- Developed an agent that derives equivalent expressions for Huawei inference operators, completing support for 20 operators.

Beijing Academy of Artificial Intelligence for Science (AISI) Jun. 2023 – Sep. 2023
High-Performance Computing Intern

- Enhanced the accuracy-testing framework for core numerical routines in ABACUS; designed accuracy tests for Simpson's integration and real spherical harmonics and integrated them into the testing workflow.

Publications and Patents

[1] **Global Scheduling of Transient Parameter Materialization for Sharded Training**
ACM Transactions on Architecture and Code Optimization (TACO). First author; CCF Rank A journal.

[2] **Memory-Safe Hierarchical Planning for Hybrid-Parallel Transformer Training**
ACM Transactions on Architecture and Code Optimization (TACO). First author; CCF Rank A journal.

[3] **Eiffel: Inferring Input Ranges of Significant Floating-point Errors via Polynomial Extrapolation**
In *Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering* (ASE 2023). First author; CCF Rank A conference.

[4] **A Decoupled Analytical Model for Tile Size Selection in Affine Programs**
ACM Transactions on Architecture and Code Optimization (TACO). Co-first author; CCF Rank A journal.

[5] **Hierarchical Search Algorithm for Error Detection in Floating-point Arithmetic Expressions**
The Journal of Supercomputing. First author; CCF Rank C journal.

[6] **Scalable Detection of Floating-point Errors via Adaptive Parallel Subdomain Search**
In *Proceedings of the 25th International Conference on Software Quality, Reliability, and Security* (QRS 2025). First author; CCF Rank C conference.

[7] Method and System for Partitioning Subintervals with Large Errors in Floating-Point Arithmetic Expressions over Specified Intervals for High-Performance Computing Function Libraries. First student inventor; patent granted.

[8] Method and Device for Evaluating Weather-Forecast Accuracy over High-Error Intervals of Fitted Expressions. First student inventor; patent application under substantive examination.

Projects

Intelligent Compiler Optimization of Parallel Strategies for Supernodes Jan. 2025 – Dec. 2026
Huawei–Hunan University Collaborative Project (TC20241115006) Student Lead

- Designed hierarchical abstractions of parallel strategies and deep compiler-co-optimization techniques for distributed training of large models.

Floating-Point Arithmetic Expression Error Detection Jan. 2023 – Dec. 2024
Open Project, State Key Laboratory of Mathematical Engineering and Advanced Computing (No. 2023B02) Student Lead

- Developed Maxfpeed, a floating-point error detection tool, and designed detection algorithms that delivered substantial gains in performance and effectiveness.

Polyhedral-Model-Based Tensor Compiler for Deep Learning Jan. 2021 – Dec. 2024
NSFC Regional Innovation and Development Joint Fund Project (No. U20A20226) Contributor

- Contributed to an automatic mixed-precision code generator that uses a polyhedral model augmented with fitted functions to identify the optimal iteration space for nested-loop programs.

XXXX Engineering Basic Math Library System Jan. 2018 – Dec. 2022
Subproject of a National Science and Technology Major Project Core Contributor

- Designed the overall testing framework for a basic math library on the domestic Sunway supercomputing platform and wrote automated test code and scripts to improve testing efficiency.

Technical Skills

- Proficient in Python and C/C++; strong foundation in object-oriented programming, data structures, and algorithms.
- Experienced with the PyTorch and MindSpore deep-learning frameworks and with Linux, Git, Docker, Make, and CMake.

Honors and Awards

- MLIR Summer School Scholarship (MLIR Foundation, May 2026)
- CSC Scholarship (China Scholarship Council, May 2026)
- PLMW Scholarship (PLDI 2025, May 2025)
- AsiaLLVM Student Scholarship (LLVM Foundation, Apr. 2025)
- Outstanding Master's Thesis Award (Information Engineering University, Oct. 2024)
- Outstanding Individual in Academic Achievement (Information Engineering University, Jun. 2024)
- National Scholarship for Graduate Students (Ministry of Education, Dec. 2023)
- First-Class Academic Scholarship (Information Engineering University, Nov. 2023)
- Second-Class Academic Scholarship (Information Engineering University, Nov. 2022)
- First-Class Academic Scholarship (Information Engineering University, Nov. 2021)
- Outstanding Student Leader (Henan University of Technology, Oct. 2018)

Academic Service

- Artifact Evaluation Committee Member, PACT 2025
- Teaching Assistant, undergraduate Compiler Principles course, Hunan University
- Invited Speaker, 20th CCF National Conference on High-Performance Computing
- Executive Committee Member, CCF Student Chapter at Hunan University